

Joel Smith

ML Systems Research Engineer

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EDUCATION

University of Liverpool

Sept 2024 – Sept 2025

MSc in Data Science & Artificial Intelligence, Distinction

University of Chester

Sept 2020 – July 2023

BSc in Economics, First Class (University Prize for Economics)

SELECTED PROJECTS

KV Cache Hinting | *Optimising memory usage for agentic workloads*

[Writeup](#)

- Designed and benchmarked a KV cache retention/eviction policy for agentic workloads served on vLLM.
- madvise-inspired block hinting (WILLNEED/DONTNEED/PAGEOUT) reordering the eviction queue against EWMA-predicted tool-call gap duration versus CPU offload/reload cost, evaluated via a custom benchmarking harness across concurrency sweeps and showing modest uplift in throughput.

Fast vLLM | *Cold-start Optimisation for vLLM Inference*

[Writeup](#)

- CRIU checkpoint/restore of a warmed vLLM process, bypassing the cold-start sequence; startup reduced from 117s to 5.8s.
- Parallel double-buffered weight loader using multi-threaded preadv to build NVMe queue depth overlapped with h2d transfer, replacing vLLM's per-tensor reload path; 6GB reload time reduced from 8s to 2.4s (NVMe floor 1.6s).

vLLM | *Open-Source Contributions*

- Merged fix for per-tensor FP8 CUTLASS kernels failing on SM 12.1 devices: replaced an overly narrow architecture guard causing a kernel trap, aligning with the existing pattern used by the blockwise FP8 path.
- Merged fix for a device-placement bug crashing MiniCPM-O on startup: a base-class override skipped a device-sync call inherited from the parent model, leaving a resampler buffer resident on CPU, causing a cross-device tensor mismatch during the forward pass.

Alchemy Runtime | *TensorRT Inference Server*

[GitHub](#)

- C++ inference server with TensorRT backend; dynamic batch queue (single drain thread, max-batch/max-wait window) with async request/response via future/promise behind a thread-pooled HTTP layer (httplib).
- On-device DDIM denoising loop minimising host-device transfers; custom CUDA kernel parallelising per-element posterior mean computation and stochastic sampling across the batch.
- Load-tested with wrk: 17 img/s peak throughput, p99 686ms at batch size 8 (RTX 5060 Ti).

Alchemy Lab | *Diffusion Model Research Framework*

[GitHub](#)

- Modular diffusion training framework built from scratch: config-driven (Hydra/OmegaConf) sweeps and ablations, checkpoint/resume for interruptible runs, seed control for reproducibility, EMA for improved sample quality, instrumented with NVTX/Nsight for bottleneck analysis.

EXPERIENCE

Artificial Intelligence Engineer

April 2026 – Present

Currys

London, UK

- Built out tabular Q&A functionality for the company-wide LLM chatbot, enabling natural language queries over structured data on a Databricks stack; also introduced file generation (pptx, docx, pdf, xlsx).
- Migrated internal chatbot codebase to Azure DevOps and implemented CI/CD pipelines for build, test, and deployment.

TECHNICAL SKILLS

Inference & Systems: LLM serving, vLLM, TensorRT, CRIU, CUDA, Nsight Systems, KV cache management, Prometheus

Machine Learning: PyTorch, DDP, Hydra/OmegaConf

Languages: Python, C++, SQL

Infrastructure: Docker, Kubernetes, Azure DevOps, Databricks, AWS, Git, Linux

ML Systems & Generative Modelling

[Blog](#)

- Deep-dives on inference performance and systems engineering.
- First-principles articles deriving generative models (GMM, VAE, diffusion) with full math and accompanying NumPy/PyTorch implementations.